

CS 4101/6103: Pattern Recognition

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Introduction

Artificial Intelligence

THE MOST EXCITING DISCIPLINE in

Today's World

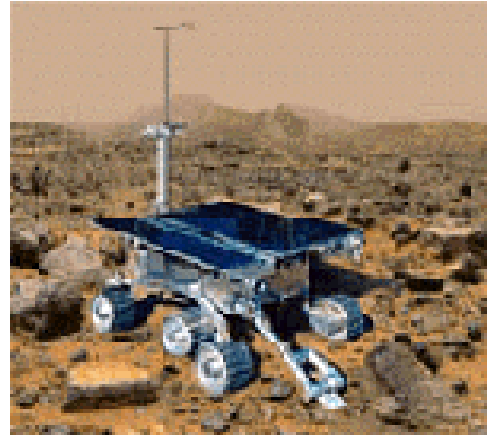
Artificial Intelligence: *Facts & Figures*

- According to Statista, revenue from the artificial intelligence (AI) software market worldwide is expected to reach **1750 billion dollars by 2032**
- As per Gartner, 37% of organizations have implemented AI in some form. The percentage of enterprises employing AI grew **270% over the past four years**
- According to Servion Global Solutions, by 2025, **95% of customer interactions will be powered by AI**

Why study AI?



Labor



Science



Search engines



Medicine/
Diagnosis



Appliances

What else?

Natural Language Question Answering



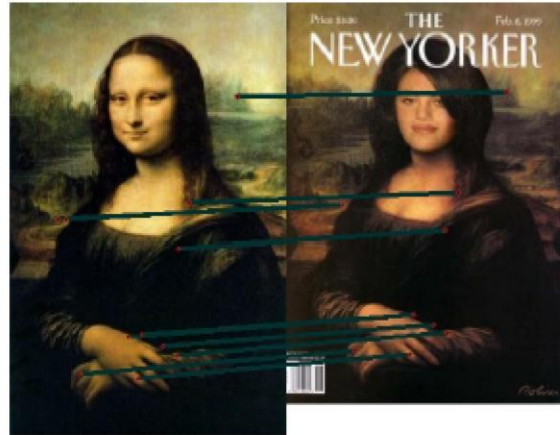
<http://aimovie.warnerbros.com>

<http://www.ai.mit.edu/projects/infolab/>

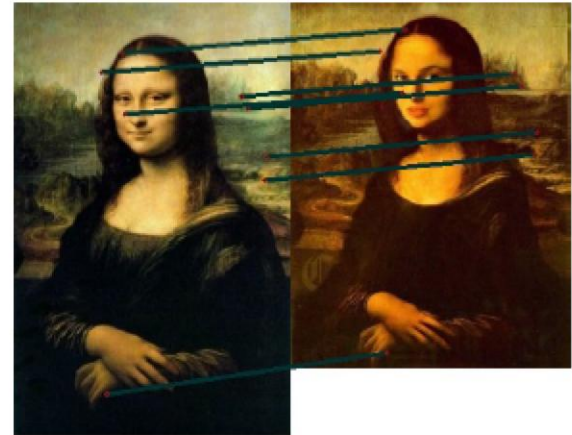
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Examples: AI at Google

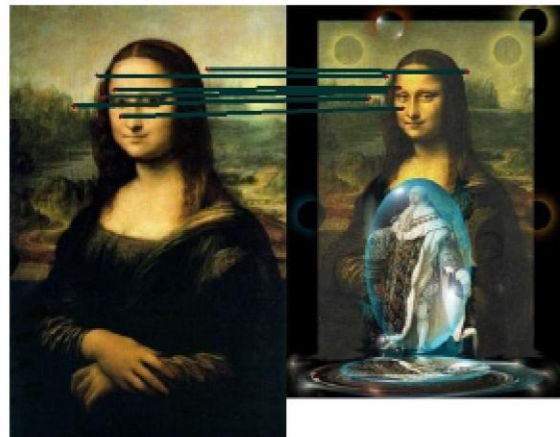
Compare low-level features



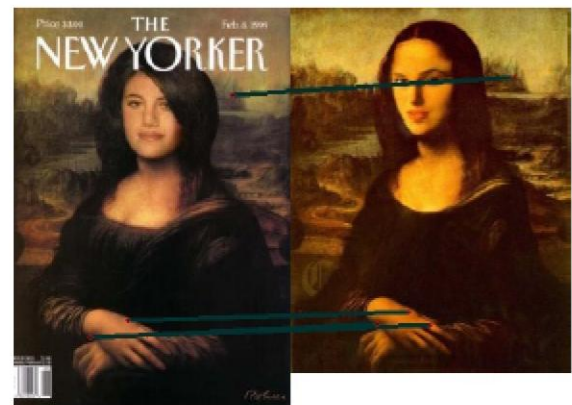
(a) A v.s. B



(b) A v.s. C

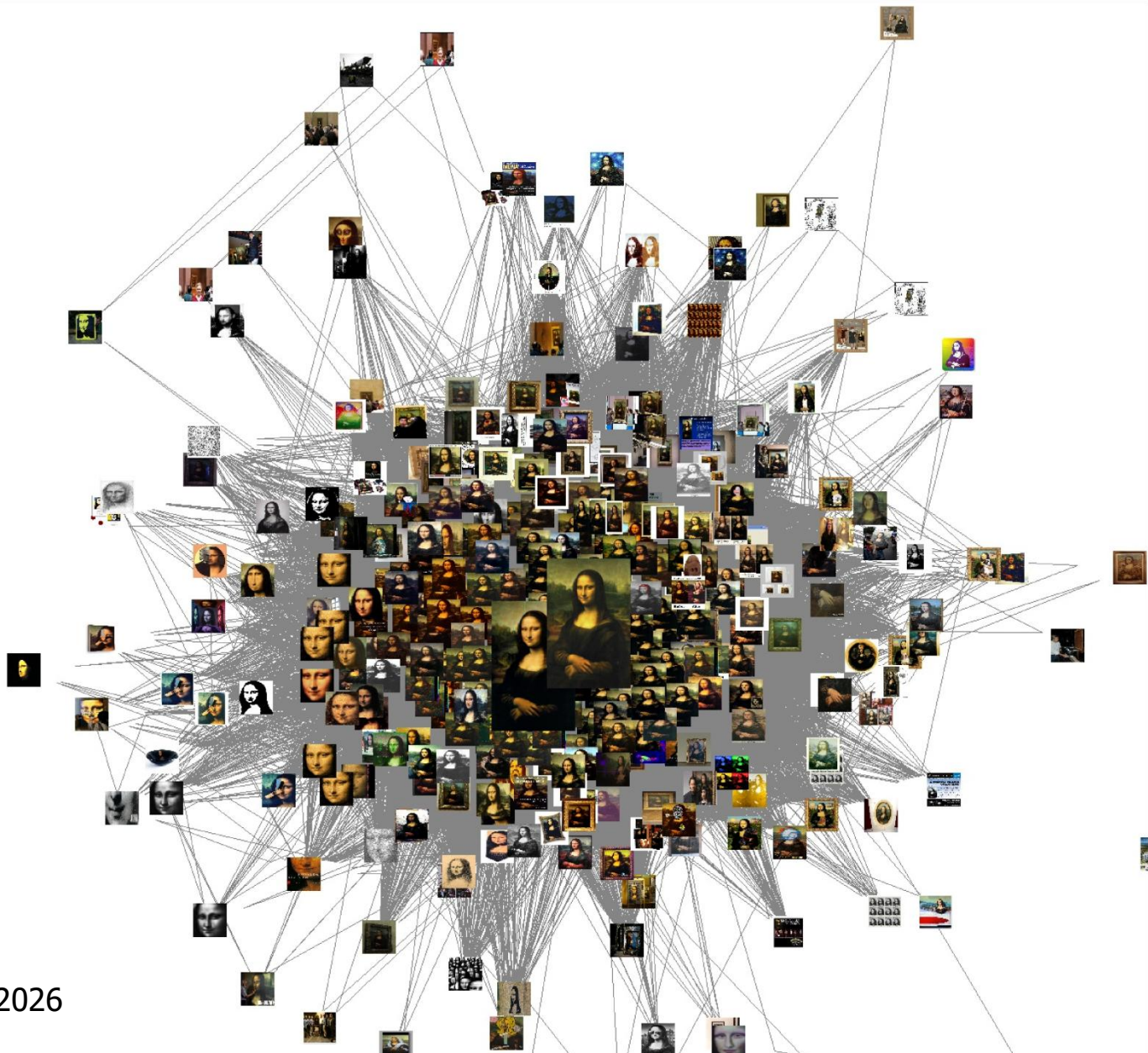


(c) A v.s. D



(d) B v.s. C

Induced Graph



8/4/2026

Route Finding

2. Turn left

210 m

3. At the roundabout, take the 2nd exit onto IIT Main Rd

250 m

Take NH922 to Anisabad Post Office Rd/Dhira Chak Main Rd/Khagaul Rd/Mithapur - Anisabad - Khagaul Rd in Anisabad, Patna

53 min (28.8 km)

4. Turn left at Singh Da Dhaba onto Binta Kanpa Rd

Pass by SURAJ Hotel (on the right in 450 m)

1.4 km

5. At Pradip Communication, continue onto Binta Rd/Bikram - Binta Rd

Continue to follow Bikram - Binta Rd

Pass by Hanuman Mandir (on the left)

2.3 km

Take IIT Main Rd to Binta Kanpa Rd

2 min (550 m)

1. Head northeast

Pass by SBI ATM (on the left)

88 m

2. Turn left

210 m

3. At the roundabout, take the 2nd exit onto IIT Main Rd

250 m

Take NH922 to Anisabad Post Office Rd/Dhira Chak Main

Rd/Khagaul Rd/Mithapur - Anisabad - Khagaul Rd in

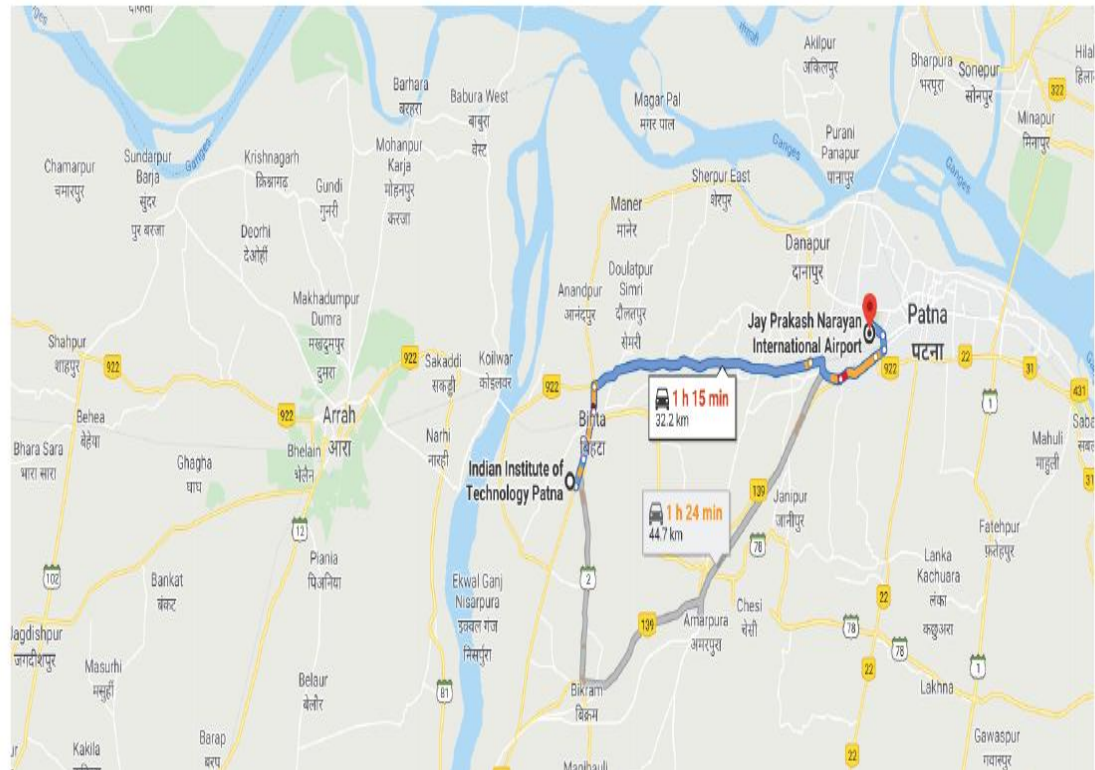
8/24/2020

Indian Institute of Technology Patna to Jay Prakash Narayan International Airport - Google Maps

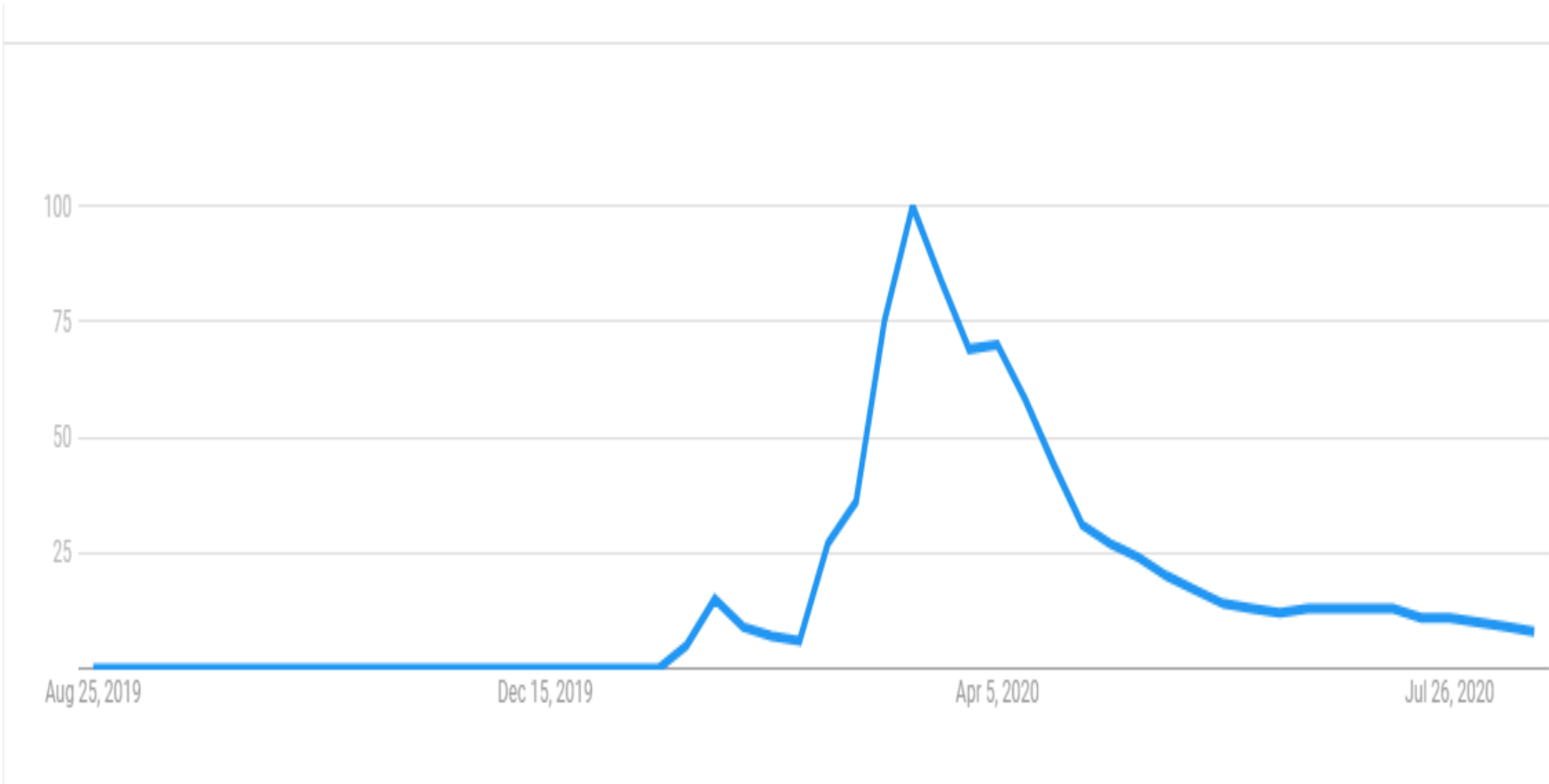


Indian Institute of Technology Patna to Jay Prakash Narayan International Airport

Drive 32.2 km, 1 h 15 min



COVID-19 Trend



8/4/2026

Artificial Intelligence: Not merely a humanoid robot

- AI, as often portrayed, in science fiction is NOT only robots or other humanoid beings
 - **who are friendly and serve humans or,**
 - **turn evil and want to kill all humans to take control of our planet**

*Fei-Fei Li, Director of Stanford AI lab, claims that the myth of the **terminator** coming next door is, in fact, a real crisis for the development of the AI field as it highlights the public misreading of the technology but also reveals the fear of what are the intentions of the people behind the technology (2018)*

- AI is, in fact, an ever-evolving term which is one of the reasons that it means very different things to different people
- Artificial Intelligence is hard to define because the field has been redefined continuously with the advances of technology and the ambiguity of what we consider as “intelligent”

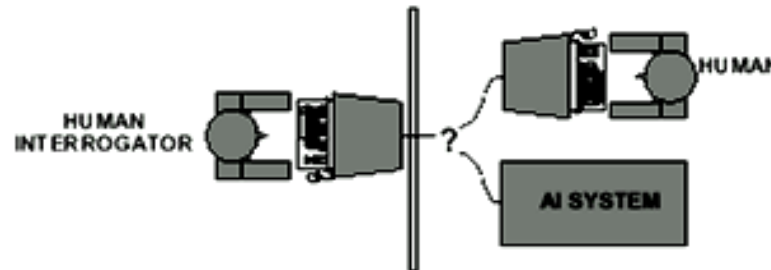
A better understanding of AI is crucial to its future development and progress

What is AI?

“The exciting new effort to make computers think ... <i>machines with minds</i>, in the full and literal sense” (Haugeland, 1985) “[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning ...” (Bellman, 1978)	“The study of mental faculties through the use of computational models” (Charniak and McDermott, 1985) “The study of the computations that make it possible to perceive, reason, and act” (Winston, 1992)				
“The art of creating machines that perform functions that require intelligence when performed by people” (Kurzweil, 1990) “The study of how to make computers do things at which, at the moment, people are better” (Rich and Knight, 1991)	“A field of study that seeks to explain and emulate intelligent behavior in terms of computational processes” (Schalkoff, 1990) “The branch of computer science that is concerned with the automation of intelligent behavior” (Luger and Stubblefield, 1993)				
Figure 1.1 Some definitions of AI. They are organized into four categories: <table><tr><td>Systems that think like humans.</td><td>Systems that think rationally.</td></tr><tr><td>Systems that act like humans.</td><td>Systems that act rationally.</td></tr></table>		Systems that think like humans.	Systems that think rationally.	Systems that act like humans.	Systems that act rationally.
Systems that think like humans.	Systems that think rationally.				
Systems that act like humans.	Systems that act rationally.				

Acting Humanly: The Full Turing Test

- Alan Turing's 1950 article *Computing Machinery and Intelligence* discussed conditions for considering a machine to be intelligent
 - “Can machines think?” $\longleftrightarrow$ “Can machines behave intelligently?”
 - The Turing test (The Imitation Game): Operational definition of intelligence

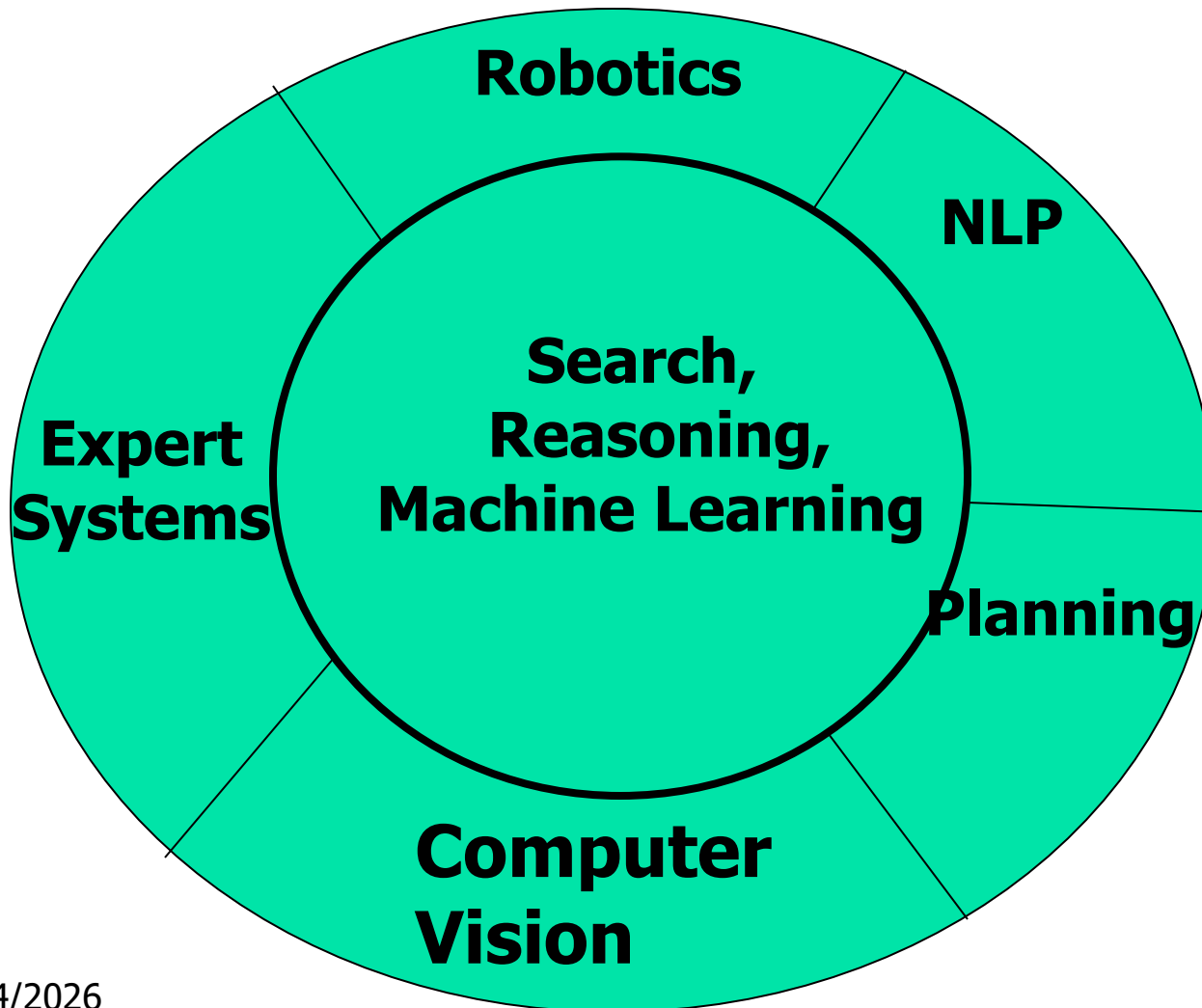


- Computer needs to possess: Natural language processing, Knowledge representation, Automated reasoning, and Machine learning
- Problem: 1) Turing test is not reproducible, constructive, and amenable to mathematic analysis. 2) What about physical interaction with interrogator and environment?
- Total Turing Test: Requires physical interaction and needs perception and actuation.

What would a computer need to pass the Turing test?

- **Natural language processing:** to communicate with examiner
- **Knowledge representation:** to store and retrieve information provided before or during interrogation
- **Automated reasoning:** to use the stored information to answer questions and to draw new conclusions
- **Machine learning:** to adapt to new circumstances and to detect and extrapolate patterns
- **Vision :** to recognize the examiner's actions and various objects presented by the examiner
- **Motor control:** to act upon objects as requested
- **Other senses:** such as audition, smell, touch, etc.

Disciplines which form the core of AI- inner circle
Fields which draw from these disciplines- outer circle



AI State-of-the-art

- Google's powerful search engines, Google's MT
- Alexa etc.
- Amazon Comprehend Medical services
- Social Networks: Facebook, Twitter etc

- **ChatGPT**

- **GPT-4**

Making sense of data



<http://youtu.be/kb7RL6b-mHE>

Defining Data

- **Data**-A collection of raw, unorganized plain facts, observations, statistics, characters, symbols, images, numbers
- **Ex:**
10, 25, ..., Patna, 10CS3002, asif@iitp.ac.in
Anything else?

Customer surveys, paper and electronic forms, CVs, and so on

- **Information** = Data + Meaning

When data is processed, evaluated, organized, structured, or presented in such a way that it becomes meaningful or helpful

Ex: avg score of a class

Big Data

Now data is Big Data!

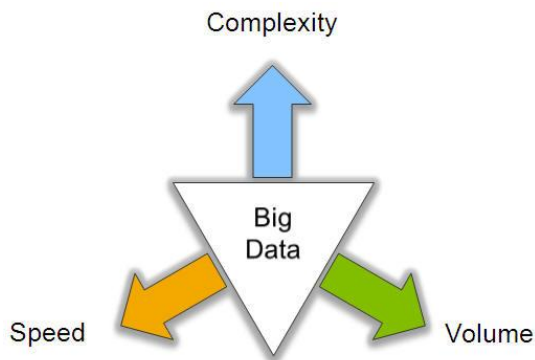
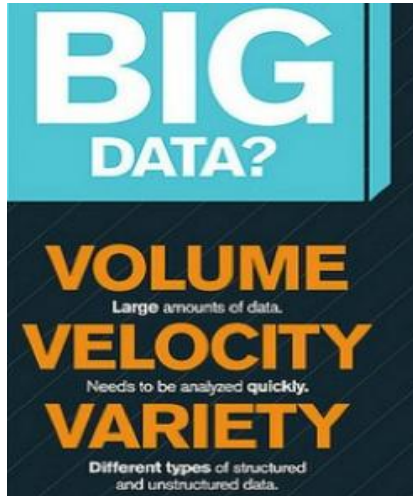
- No single standard definition!
- 'Big-data' is similar to 'Small-data', but bigger
...but having data bigger consequently requires different approaches
 - techniques, tools and architectures

...to solve: new problems

...and, of course, in a better way

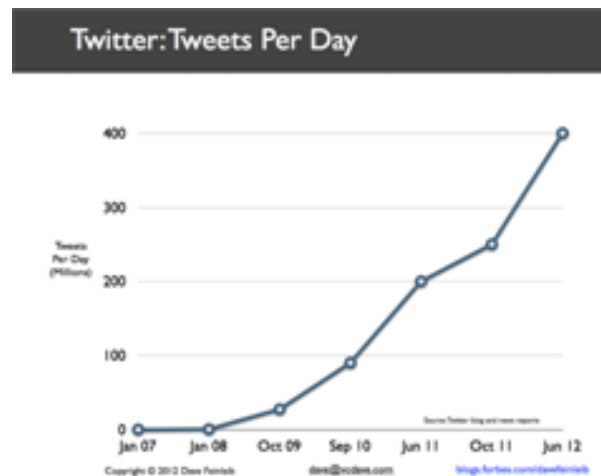
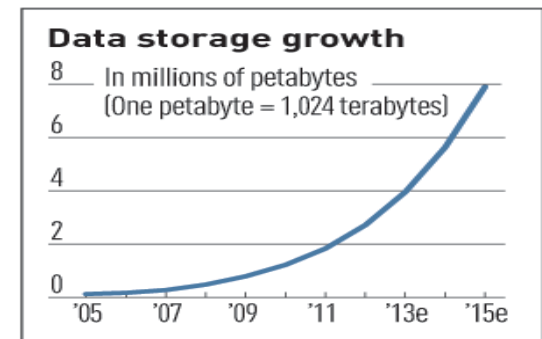
Big data is data whose scale, diversity, and complexity require new architecture, techniques, algorithms, and **analytics** to manage it and extract value and hidden knowledge from it...

Characteristics of Big Data: V3



V3 : V for Volume

- Volume of data, which needs to be processed is increasing rapidly
 - More storage capacity
 - More computation
 - More tools and techniques

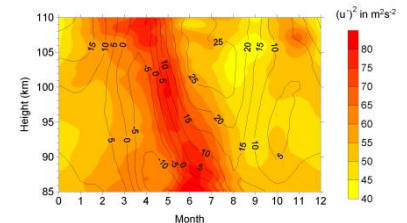
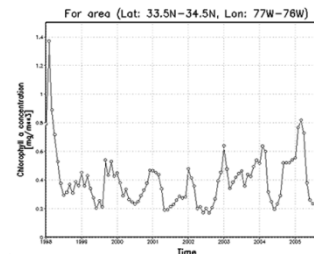
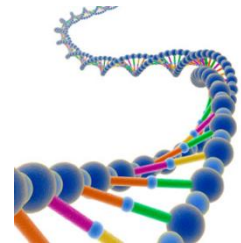
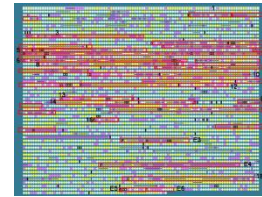


Exponential increase in collected/generated data

V3: V for Variety

- Various formats, types, and structures
 - Text, numerical, images, audio, video, sequences, time series, social media data, multi-dimensional arrays, etc...
- Static data vs. streaming data
- A single application can be generating/collecting many types of data

To extract knowledge → all these types of data need to be linked together



V3: V for Velocity

- Data is being generated fast and need to be processed fast
 - For time-sensitive processes such as catching fraud, big data must be used as it streams into your enterprise in order to maximize its value
 - Scrutinize 5 million trade events created each day to identify potential fraud
 - Analyze 500 million daily call detail records in real-time to predict customer churn faster



Big data: Blessing or curse?

Information is the main treasure of humankind

Without efficient management, such a treasure becomes useless:
the more we have, the less we can use!

Web is Enormous

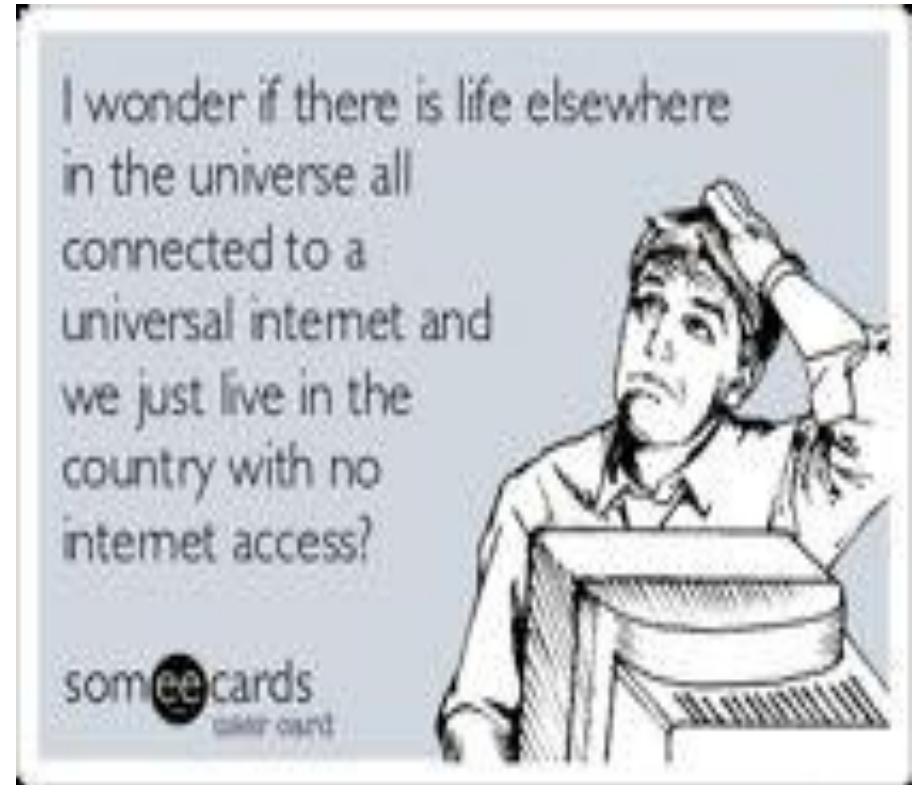
Web: connecting people

- The potential for Knowledge sharing today is unmatched in history
- Never before have so many knowledgeable people been connected

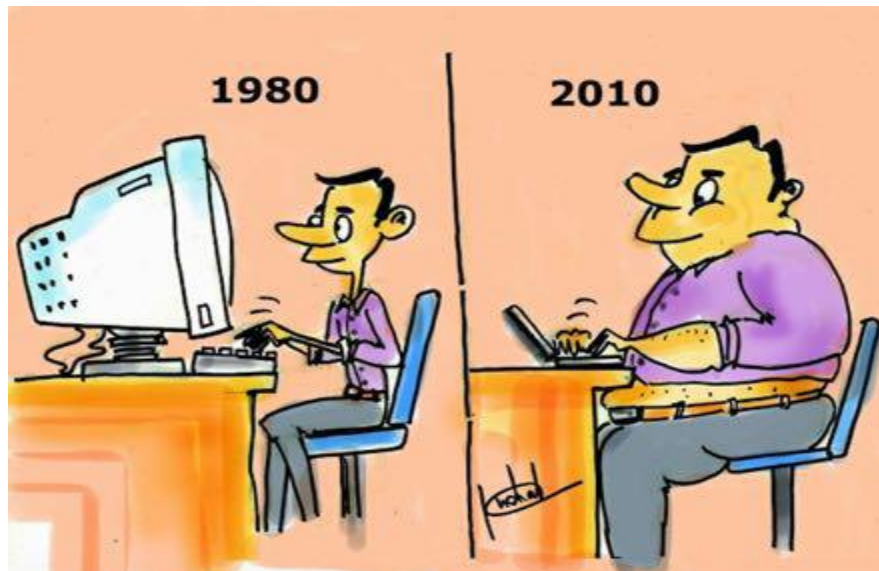


To be or not to be connected?

***Being connected is good
but being disconnected for
the past millions years was
the main reason behind our
cultural diversity***



Being disconnected!



Machine Learning

- **Machine learning**: how to acquire a model on the basis of data / experience?
 - Learning parameters (e.g. probabilities)
 - Learning structure (e.g. BN graphs)
 - Learning hidden concepts (e.g. clustering)

Machine Learning

- Unsupervised Learning
 - No feedback from teacher; detect patterns
- Reinforcement Learning
 - Feedback consists of rewards/punishment
- Supervised Learning
 - Examples of correct answers are given
 - Discrete answers: *Classification*
 - Continuous answers: *Regression*

AI & ML: From Past to Present

■ Classical ML algorithms

- Used extensively (Rule-based, Naïve Bayes, HMM, MaxEnt, CRF, SVM, Logistic Regression etc.)
- Requires handcrafting of features
- Time-consuming
- Curse of dimensionality

■ Deep Learning algorithms

- No feature engineering
- Success of distributed representations (especially for Natural Language Processing System)

Top-10 Classical ML/Deep Learning algorithms : ICDM-2006

- C4.5 (Decision Tree)
- k-Means
- SVM
- Apriori
- EM
- PageRank
- AdaBoost
- kNN
- **Naive Bayes**
- CART

Top 10 data mining algorithms. Knowl Inf Syst (2008) 14:1–37 DOI 10.1007/s10115-007-0114-2

AI & ML: From Past to Present

- **Some recent developments in Deep Learning**
 - The rise of distributed representations (e.g., Word2vec, GLOVE, ELMO, BERT etc)
 - Convolutional, recurrent, recursive neural networks, Transformer, Reinforcement learning
 - Unsupervised sentence representation learning
 - Combining deep learning models with memory-augmenting strategies
- Explainable AI

AI: Future

- Artificial General Intelligence :
able to exhibit human intelligence
- Artificial Super Intelligence :
surpasses human intelligence in many aspects

from creativity to general wisdom to problem-solving —
will require machines to experience consciousness

Recognition for Modern AI

*Yoshua Bengio, Geoffrey Hinton, and Yann LeCun
received the*

*Turing Award-2018 (equivalent to Nobel Prize of
Computing)*

- *for Modern AI (specifically for deep learning
research)*

Bengio- University of Montreal

Hinton- University of Toronoto & Google

LeCun- Facebook's chief AI scientist and a professor at
NYU

Goal of Teaching the course

- **Concept building**: firm grip on foundations, clear ideas
- **Coverage**: grasp of good amount of material, advances
- **Inspiration**: get the spirit of ML, motivation to take up further work

Course Contents

- **Introduction to Pattern Recognition:** Learning paradigms, Supervised and unsupervised learning
- **Traditional Supervised paradigms:** Decision Trees, Bayesian Classification, Rule-based learning
- **Ensemble Learning:** Bagging, Boosting
- **Graphical Models:** Hidden Markov Models, Intro to MaxEnt and CRF
- **Neural Networks and Deep Learning:** Perceptron, Backpropagation, Recurrent Neural Network, LSTM, GRU, Encoder-Decoder, Attention, Autoencoder, Transformer, GAN
- **Unsupervised Learning and Feature selection:** PCA, LDA, Evolutionary algorithm for feature selection, k-means clustering, k-medoid

Recent Trends (Theory and Application)

Evaluation

MSE- 25%

ESE- 40%

Assignments- 35%

Resources

- T. Mitchell. Machine Learning. McGraw-Hill, 1997.
- Christopher Bishop. Pattern recognition and machine learning. Springer Verlag, 2006.
- Hastie, Tibshirani, Friedman. The elements of Statistical Learning Springer Verlag.
- Probability, Random Variables and Stochastic processes by Papoulis and Pillai, 4th Edition, Tata McGraw Hill Edition.
- A. K. Jain and R. C. Dubes. Algorithms for Clustering Data. Prentice Hall, 198815.

Thank you for your attention!